Guido Gagliardi

Personal Information

Guido Gagliardi is a PhD student pursuing a double degree in the International PhD Program in Smart Computing at the University of Pisa, Florence, and Siena (Italy) and the Doctoral Program in Engineering Science at KU Leuven (Belgium). With his research, he is collaborating with national (FAIR) and international (EXPERIENCE) research projects.

His research is focused on developing novel explainable artificial intelligence techniques for analyzing physiological signals to enhance the decision-making process in clinical scenarios. His proposition is that the utilization of artificial intelligence approaches, such as neural networks, presents a significant opportunity for advancing human understanding of brain functionality and, consequently, improving overall quality of life.

His dream is to pursue a research career wherein I employ explainable artificial intelligence (XAI) to investigate the human brain through neural networks in clinical scenarios. He aims to cultivate and refine my skills and techniques in diverse and authentic contexts, emphasizing engagement with multidisciplinary and international academic environments. This trajectory is exemplified by the choice of a joint degree program and is a trajectory he seeks to continue exploring beyond the boundaries of Europe.

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Date of birth:	**25 September 1996**
Birthplace:	**Pisa (PI)**
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Education

Nov. 2021—Pres. **double degree of Doctor of Philosophy (PhD.)**

Course	**Smart Computing** **Doctoral Programme in Engineering Science**
Institutes	**University of Pisa, Florence, and Siena (Italy)** **KU Leuven (Belgium)**
Expected Grad. Date	**Oct. 2024**

INTERNATIONAL SCIENTIFIC JOURNAL PUBLICATIONS (WITH PEER REVIEW)

Title	**Concept-wise granular computing for explainable artificial intelligence**
Contributors	Alfeo, A.L.; Cimino, M.G.C.A.; Gagliardi, G.
DOI	*https://doi.org/10.1007/s41066-022-00357-8*
Journal	Granular Computing
Title	**Improving emotion recognition systems by exploiting the spatial information of EEG sensors**
Contributors	*Guido Gagliardi; Antonio Luca Alfeo; Vincenzo Catrambone; Diego Candia-Rivera; Mario G. C. A. Cimino; Gaetano Valenza*
DOI	*https://doi.org/10.1109/ACCESS.2023.3268233*
Journal	**IEEE Access**
Title	**Emotion-Specific Brain-Heart Interplay Saliency Maps using eXplainable Artificial Intelligence**
Contributors	*Guido Gagliardi; Antonio Luca Alfeo; Vincenzo Catrambone; Mario G. C. A. Cimino; Maarten De Vos; Gaetano Valenza*
DOI	NA
Journal	**Submitted to an international journal**
Title	**Recognizing bearings' degradation stage by using multi-modal autoencoder to learn features from different time series**
Contributors	*Antonio Luca Alfeo; Mario G. C. A. Cimino; Guido Gagliardi*
DOI	NA
Journal	**Submitted to an international journal**

INTERNATIONAL CONFERENCE PUBBLICATIONS (WITH PEER REVIEW)

Title	**Matching the expert's knowledge via a counterfactual-based feature importance measure**
DOI	**NA – Submitted to an international conference**
Contributors	*Antonio Luca Alfeo; Mario G. C. A. Cimino; Guido Gagliardi*
Title	**Fine-grain Emotion Recognition using Brain-Heart Interplay measurements and eXplainable Convolutional Neural Networks**
DOI	*https://doi.org/10.1109/NER52421.2023.10123758*
Contributors	*Guido Gagliardi; Antonio Luca Alfeo; Vincenzo Catrambone; Mario G. C. A. Cimino; Maarten De Vos; Gaetano Valenza*

Conference	**IEEE N.E.R. 2023**
Title	**Automatic feature extraction for bearings' degradation assessment using minimally pre-processed time series and multi-modal feature learning**
Contributors	Alfeo, A.L.; Cimino, M.G.C.A.; Gagliardi, G.
DOI	https://doi.org/10.5220/0011548000003329
Conference	**Proceedings of the 3rd International Conference on Innovative Intelligent Industrial Production and Logistics (IN4PL 2022)**
Title	**Improving an Ensemble of Neural Networks via a Novel Multi-class Decomposition Schema**
Contributors	Alfeo, A.L.; Cimino, M.G.C.A.; Gagliardi, G.
DOI	https://doi.org/10.5220/0011547900003332
Conference	**Proc. of the 14th International Joint Conference on Computational Intelligence – Vol. 1: NCTA**

PHD Courses

5 CFU	**English for Research and Academical Purposes**
3 CFU	**Theoretical fundamentals of Neural Networks**

Teaching

A.A. 2022-2023 — **Lab. Sessions Course: Biomedical Signal Processing, ESAT, KU Leuven**

Advised Thesis

G. Cancello Tortora, "Analyzing brain data for robust emotion recognition via conceptual decomposition based on autoencoder", MS in Artificial Intelligence and Data Engineering, University of Pisa, a.y. 2022-23.

F. Marabotto, "Explainable emotion recognition via a novel loss function based on informed contrastive learning", MS in Artificial Intelligence and Data Engineering, University of Pisa, a.y. 2022-23.

L. Turchetti, "Sleep stage recognition supported by instances-based explanation via contrastive learning", MS in Artificial Intelligence and Data Engineering, University of Pisa, a.y. 2022-23.

P. Calabrese, "Explainable Artificial Intelligence Approaches for Industrial Data Science", MS in Artificial Intelligence and Data Engineering, University of Pisa, a.y. 2022-23

M. Martorana, A Novel Feature Importance Measure To Explain The Quality Level Prediction In - Smart Manufacturing, MS in Artificial Intelligence and Data Engineering, University of Pisa, a.y. 2021-22.

N. Mota, "Concept-wise architecture with topology learning for explainable emotion classification" MS in Computer Engineering, a.y. 2021-22.

F. Ritorti, Design and testing of a loss function for distance-based representation learning to recognize emotions via EEG data, MS in Artificial Intelligence and Data Engineering, a.y. 2021-22.

Sep. 2019—Sep. 2021 Master's degree

Course	**Artificial Intelligence and Data Engineering**
Institute	**University of Pisa**
Graduation Mark	**110 *cum Laude***
Graduation Date	**24/09/2021**
Exams' mark weighted average	**30.0**

Thesis Title — **A novel multimodal feature learning architecture for explainable affective computing**

Thesis Supervisors	**Mario G.C.A. Cimino, Gigliola Vaglini, Antonio Luca Alfeo**
Thesis Abstract	Development of AI architectures apt to recognize emotional states from electroencephalogram and other physiological signals and to provide insights on how the output is established.

Master's degree Exams

Code	Title	Mark/30
878II	DATA MINING AND MACHINE LEARNING	27
883II	LARGE-SCALE AND MULTI-STRUCTURED DATABASES	30
876II	CLOUD COMPUTING	30
875II	BUSINESS AND PROJECT MANAGEMENT	30
696AA	OPTIMIZATION METHODS AND GAME THEORY	27
891II	ROBOTICA E MACCHINE INTELLIGENTI	**30 cum Laude**
882II	INTERNET OF THINGS	30
886II	MULTIMEDIA INFORMATION RETRIEVAL AND COMPUTER VISION	30
888II	PROCESS MINING AND INTELLIGENCE	**30 cum Laude**
910II	INDUSTRIAL APPLICATIONS	30
877II	COMPUTATIONAL INTELLIGENCE AND DEEP LEARNING	30
893II	SYMBOLIC AND EVOLUTIONARY ARTIFICIAL INTELLIGENCE	**30 cum Laude**

AVERAGE SCORE	30.0

Class Project and Contributions

Title	**Heartbeat anomalies detection using a Data Mining approach**
Course	Data Mining and Machine Learning
Abstract	The project investigated the usage of Data Mining approaches to face arrhythmia detection on 5 classes of arrhythmia on the MIT-BIH dataset.
Title	**Medical image analysis using CNN**
Course	Computational Intelligence and Deep Learning
Abstract	The project investigated the possibility to use Convolutional Neural Networks and Siamese models to perform abnormalities detection in medical images such as mammography for cancer detection. **I proposed and constructed a Generative Adversarial Network approach** to oversample the dataset to prevent overfitting.
Title	**A Deep Reinforcement Learning approach to train multi-agent robot system for formation control**
Course	Robotica e Macchine Intelligenti
Abstract	The project investigated the use of Deep Reinforcement Learning to train a multi-agent robot network, in a decentralized fashion.
Title	**Interior Point Method improvements using sparse matrixes**
Course	Symbolic and evolutionary Artificial Intelligence
Abstract	The project investigated possibility to improve the speed performances of the Interior Point Method algorithm in a multi-objective environment using sparse matrixes to solve the Karush Kuhn Tucker system related to the method.

Sep 2015—Feb 2019 **Bachelor's Degree**

Course	**Computer Engineering**
Institute	**University of Pisa**
Graduation Mark	**105/110**

Thesis Title:	**Novelty data detection methods**
Thesis Supervisors	**Francesco Marcelloni, Alessio Bechini**
Thesis Abstract	AI deployment on outlier detection task, based on two algorithms of LE (learning entropy) and ELBND (error and learning-based novelty detection).System testing with both general stream of data and in presence of concept drift.

Sep 2010—July 2015 **High School Diploma**

Institute	**Liceo Scientifico Ulisse Dini, Pisa (PI), Italy (IT)**
Mark	**87/100**

Skills and Ability

MOTHER TONGUE	ITALIAN
OTHER LANGUAGES	**ENGLISH**
Level	**C1**

TECHNICAL SKILLS

Artificial Intelligence, eXplainable Artificial Intelligence, Brain Signal Analysis, Biomedical Data Analysis, Deep Learning, Data Mining, Machine Learning, Robotics, Real-Time Programming, Internet of Things, Networking.

C++, Python, Pytorch, Tensorflow, Captum, MNE, Julia, ROS, Java, UML.

RELATIONAL SKILLS

Team leadership abilities acquired both through academic group projects and by participating in University of Pisa Radio, RadioEco as President and Station Manager. Aptitude to work in well competitive and stressful scenarios with close deadlines as proven during my academic career.